keerthanakuppan15-jpg / Univariate-Linear-Regression

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Implementation of Univariate Linear Regression

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Public repository · Forked from etjabajasphin/Univariate-Linear-Regression

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This branch is 1 commit ahead of `etjabajasphin/Univariate-Linear-Regression:main` .

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README.md	Update README.md	now
eq1.jpg	experiment added	5 years ago
eq2.jpg	experiment added	5 years ago
input.jpg	experiment added	5 years ago

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Implementation of Univariate Linear Regression

Aim:

To implement univariate Linear Regression to fit a straight line using least squares.

Equipment's required:

1. Hardware – PCs

2. Anaconda – Python 3.7 Installation / Moodle-Code Runner

Algorithm:

1. Get the independent variable X and dependent variable Y.
2. Calculate the mean of the X -values and the mean of the Y -values.
3. Find the slope m of the line of best fit using the formula.

$$m = \frac{\sum_{i=1}^{n}(x_i - \overline{X})(y_i - \overline{Y})}{\sum_{i=1}^{n}(x_i - \overline{X})^2}$$

4. Compute the y -intercept of the line by using the formula:

$$b = \overline{Y} - m\overline{X}$$

5. Use the slope m and the y -intercept to form the equation of the line.
6. Obtain the straight line equation Y=mX+b and plot the scatterplot.

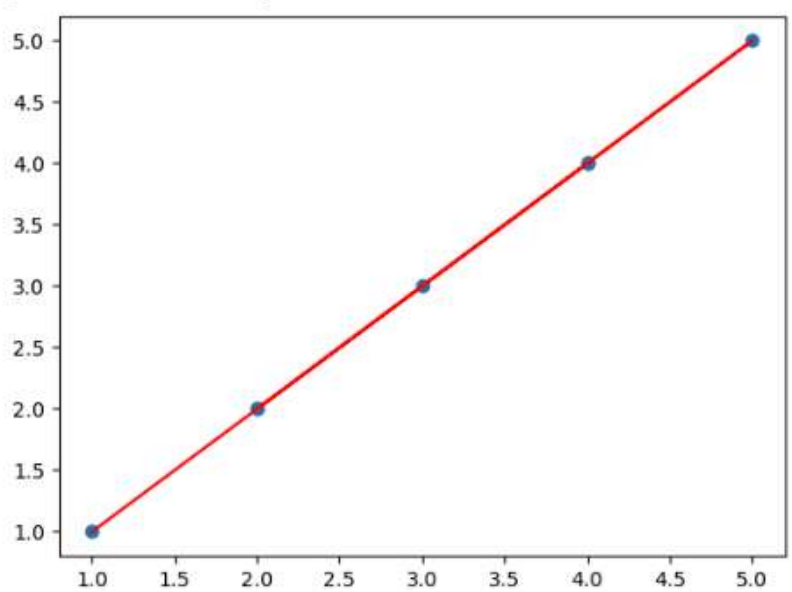
Program

```
Program to develop to implement univariate Linear Regression to fit a straight line using
least squares.

Developed by : Keerthana k
Register no : 212225230137

import numpy as np
import matplotlib.pyplot as plt
X= np.array([0,1,2,3,4,5,6,7,8,9])
Y= np.array([1,3,2,5,7,8,8,9,10,12])
plt.scatter(X,Y)
plt.show()
X_Mean=np.mean(X)
Y_Mean=np.mean(Y)
num=0
den=0
for i in range(len(X)):
    num+=(X[i]-X_Mean)*(Y[i]-Y_Mean)
    den+=(X[i]-X_Mean)**2

m=num/den
b=Y_Mean-(m*X_Mean)
print(f"Slope : {m}\nIntercept : {b}")
Y_Pred=(m*X)+b
print(f"Predicted values are : \n{Y_Pred}")
plt.scatter(X,Y,color='Red')
plt.plot(X,Y_Pred,color='Blue')
plt.show()
```

Output

```
developed by : keerthana k
register no : 212225230137

import numpy as np
import matplotlib.pyplot as plt

X = np.array(list(map(float, input().split())))
Y = np.array(list(map(float, input().split())))

m = np.sum((X - X.mean()) * (Y - Y.mean())) / np.sum((X - X.mean())**2)
c = Y.mean() - m * X.mean()

Y_pred = m * X + c

print(m, c)
print(Y_pred)

plt.scatter(X, Y)
plt.plot(X, Y_pred, 'r')
plt.show()
```

```
 1 2 3 4 2 4 5 4
 1 2 3 4 2 4 5 4
1.0 0.0
[1. 2. 3. 4. 2. 4. 5. 4.]
```



Result

Thus the univariate Linear Regression was implemented to fit a straight line using least squares.

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